



Jialing Dai

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Research Interests

Combinatorial optimization and approximation algorithms; submodular and non-submodular optimization; **graph algorithms and network science**; opinion dynamics and influence diffusion; robust coverage and resource allocation; **graph neural networks and reinforcement learning for combinatorial decision-making**.



Research Profile

- **Research focus:** Rigorous and scalable models and algorithms for decision-making on complex networks, particularly under group effects, competition, uncertainty, and signed interactions.
- **Methodological strengths:** Complexity and approximation analysis, sampling and graph algorithms, and learning-assisted combinatorial optimization.
- **End-to-end research execution:** Lead research projects from problem formulation and theoretical analysis through algorithm design, implementation, computational experiments, visualization, and manuscript preparation.
- **Track record:** Four first-author papers form the core research line; seven peer-reviewed papers have been published or accepted, including five with first- or corresponding-author roles.



Education and Research Appointments

**Ph.D. Candidate in Computer Application Technology (Computer Science and Technology),
University of Chinese Academy of Sciences (UCAS)**

Beijing, China
Sep. 2020 - Present

- Advisor: Prof. Jianming Zhu; GPA: 3.83/4.00.
- **Dissertation:** Formal Modeling and Optimization Algorithms for Group Opinion Dynamics and Influence-Related Decision Problems.
- Planned dissertation defense: **Aug. 13, 2026**; degree expected in **late Sep. 2026**.

Visiting Researcher, Department of Computer Science, The University of Texas at Dallas

Richardson, TX, USA
Jan. 2025 - Jan. 2026

- Faculty hosts: Profs. Weili Wu and Ding-Zhu Du.
- Collaborative research on graph algorithms, robust coverage, resource diffusion, and transferable networked decision models across changing application contexts.

**Bachelor of Management in Information Management and Information Systems, Beijing
University of Chemical Technology (BUCT)**

Beijing, China
Sep. 2016 - Jun. 2020

- GPA: 92.68/100; ranked 1st in major.
- Interdisciplinary training in information systems, data analytics, operations research, and management.



Publications and Manuscripts

Peer-Reviewed Journal Publications

- **Jialing Dai**, Yisheng Zhou, Yefeng Sun, Jianming Zhu, Weili Wu. "Unified Group-Aware Influence Maximization with Generalized Deep Reinforcement Learning." *IEEE Transactions on Networking (JCR Q1)*, 2026. **First author**. doi:10.1109/TON.2026.3711960
- **Jialing Dai**, Jianming Zhu, Guoqing Wang. "Competitive net influence maximization on intergroup debate effect." *Information Sciences (JCR Q1)*, vol. 680, article 121139 (2024). **First author**. doi:10.1016/j.ins.2024.121139
- **Jialing Dai**, Jianming Zhu, Guoqing Wang. "Opinion influence maximization problem in online social networks based on group polarization effect." *Information Sciences (JCR Q1)*, vol. 609, pp. 195-214 (2022). **First author**. doi:10.1016/j.ins.2022.07.086
- **Jialing Dai**, Jianming Zhu, Guoqing Wang, Jun Huang. "Net positive information diffusion activity maximizing in signed online social networks." *Chinese Journal of Management Science*, vol. 33, no. 3, pp. 139-150 (2025) [in Chinese]. **First author**. doi:10.16381/j.cnki.issn1003-207x.2022.0173
- Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "PROMO-Orchard: A Robust Semantic-Enhanced Framework for Dual-Robot Simultaneous Localization and Mapping Merging in Orchard." *Journal of Field Robotics*, **published online**, 14 Aug. 2026. DOI: 10.1002/rob.70323.

Peer-Reviewed Conference Publications

- Ke Su, **Jialing Dai (corresponding author)**, Weili Wu. "Scenario-Robust Checkpoint Cover for AI-Generated Content Provenance." *AAIM 2026*, **accepted**; Springer LNCS.
- Yisheng Zhou, **Jialing Dai**, Dongyu Mao, Ke Su, Weili Wu. "Modeling Organizational Performance through Competitive Resource Diffusion." *AAIM 2026*, **accepted**; Springer LNCS.

- Yefeng Sun, Liang Gong, **Jialing Dai**, et al. "S2BEV: Lightweight, Robust, and Precise SLAM-Oriented Segmentation Bird Eye's View Mapping Approach." *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 13160-13165 (2025). doi:10.1109/ICRA55743.2025.11127686

Revised Manuscripts Submitted

- **Jialing Dai**, Ke Su, Jianming Zhu, Weili Wu, Ding-Zhu Du, Peikun Ni. "Influence Maximization in Online Social Networks: Models, Algorithms, and Trustworthy Decision-Making." *Expert Systems with Applications (JCR Q1)*, **revised manuscript submitted**.
- Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "CaMT: Class-aware Multi-level Training Distillation for Lightweight Semantic Segmentation in Apple Orchard." *Artificial Intelligence in Agriculture*, **revised manuscript submitted**.

Manuscript Under Double-Anonymous Review

- Manuscript on multi-robot coordination and cloud-based control; title and author list withheld while **under double-anonymous review**.



Selected Research Contributions

Group-aware influence maximization and UG-DCGNN

IEEE T-ON
2026

- Proposed a **unified representation for group-aware diffusion mechanisms**, covering echo-chamber effects, topic-aware exposure, homophily, and related group channels.
- Analyzed **non-submodularity, non-supermodularity, and constant-factor inapproximability** regimes, clarifying when classical influence-maximization guarantees no longer apply.
- Developed **UG-DCGNN**, a coupled **graph-neural-network** and **DDQN solver** that learns marginal-gain prediction and reduces dependence on inference-time Monte Carlo simulation.
- Designed ablation and sensitivity studies for group-channel aggregation, multi-step seed selection, learned marginal gains, and robustness across network settings.
- Generalized the formulation to decision settings where group structure changes the objective landscape.

Competitive net influence under intergroup debate

Information Sciences
2024

- Developed the **IDIC model** to capture cross-group opinion interactions through a hypergraph coupling between group-consensus nodes and member nodes.
- Formalized a competitive net-influence objective and proved **NP-hardness and non-submodularity**.
- Introduced **HCIC** with type-dependent activation probabilities and constructed **submodular upper and lower bounds** for the original objective.
- Designed **heterogeneous competitive influence sampling**, greedy seed selection with sampling refinement, tailored data structures, lazy gains, pruning, and reverse-sampling evaluation.
- Used a **sandwich framework** with submodular upper- and lower-bound objectives to obtain **data-dependent approximation guarantees** for the original non-submodular objective; showed that stronger group cohesion can drive explosive growth in debate-related information activity.

Opinion influence under group polarization

Information Sciences
2022

- Built a real-world Zhihu dataset comprising **50 original posts and 127,178 comments across multiple reply levels (including nested replies)**, preserving the hierarchical discussion structure for empirical detection and operational modeling of group polarization.
- Proposed an enhanced independent-cascade model with preference updates to capture shock and aversion when persuasive exposure meets large preference gaps.
- Designed **GEIM** with structure-aware candidate selection and seed adjustment; connected preference dynamics to algorithmic seed-selection decisions.
- Derived and implemented update rules for signed preference change and analyzed how initial preference, group size, and opposing opinions shape influence outcomes.
- Showed that for extreme groups, **wider propagation can backfire** and reduce the objective rather than improve it.

Signed-network activity maximization

Chinese Journal of
Management Science
2025

- Shifted signed diffusion from node coverage to **edge-level net-positive activity**, explicitly modeling positive and negative interactions.
- Reformulated a non-submodular objective through a **path-level coverage surrogate** and solved it through **maximum-coverage greedy optimization with CELF acceleration**.
- Analyzed positive and negative propagation paths to distinguish reach from beneficial activity.
- Produced signed-aware diagnostics and ablation studies showing that **reaching more users is not always beneficial** under negative word-of-mouth.

Collaborative extensions

AAIM 2026
accepted papers

- Contributed to scenario-robust checkpoint-cover modeling.
- Supported extensions to uncertain generation and propagation scenarios.
- Contributed to competitive resource-diffusion modeling.
- Applied networked decision methods to organizational resource allocation.

Selected Projects and Academic Experience

2025 International Young Partners Program

Proposal Lead & Project
Coordinator
Dec. 2025 - May 2026

- Led **proposal development and end-to-end delivery** for the China Science and Technology Development Foundation's 2025 International Young Partners Program, including a **five-day academic exchange** in China with **Dr. Hongmin (William) Du** of UMass Boston.
- Designed the academic agenda; organized seminars and research discussions; and completed project reporting and closeout.
- Exchange topics included large language models, cybersecurity information disclosure, and data-driven risk analytics.

Risk and Safety Analytics for Power-Grid Construction

Student Team Lead
Oct. 2022 - Dec. 2022

- Led a **6-member student team** under faculty supervision in an applied analytics collaboration with State Grid Henan Electric Power Research Institute.
- Built a hierarchical safety taxonomy and a **1,498-case labeled dataset** from accident and safety-management materials.
- Ran **association-rule and clustering analyses**, summarized risk patterns, and drafted targeted management recommendations for safety scenarios.

Hazard-Source Tracing with Fault-Tree Modeling

Student Team Co-Lead
May 2022 - Apr. 2023

- Co-led the student research team under faculty supervision; developed a hazard-source tracing method based on fault-tree analysis and completed multi-case studies.
- Contributed major sections to the final report, translating heterogeneous incident evidence into structured models and policy-facing analysis.

Book chapter, translation, and technical writing contributions

Advisor-Supervised Contributor
2023 - 2024

- Drafted Chapter 7 sections and figures for a book on online social-network information dissemination; prepared camera-ready materials per house style.
- Contributed LaTeX-based translation work for Introduction to Operations Research, 11th ed., including Integer Programming and Heuristics chapters.
- Strengthened technical communication across modeling, algorithms, figure preparation, translation, and publication-ready formatting.

Selected undergraduate modeling work

Undergraduate
2017 - 2020

- Implemented an improved NSGA-II for multi-objective flexible job-shop scheduling with niche retention, elitism, adaptive operators, greedy initialization, Gantt diagnostics, and Pareto-front evaluation.
- Modeled periodic rail-guided vehicle (RGV) scheduling with improved genetic algorithms covering tool allocation, vehicle sequencing, and batch-size decisions; compared against a vanilla GA baseline.
- Contributed to mathematical modeling contests using cellular automata for SIS-style opioid spread modeling and SVM-based traffic-congestion prediction.

Selected undergraduate systems and research-visit work

Undergraduate
2018 - 2020

- Served as student project lead for a Hadoop Data Wall service-outsourcing contest entry; configured pseudo-distributed Hadoop, built ETL into MySQL, and delivered an interactive dashboard demo.
- Completed a National University of Singapore research-visit project involving the setup and evaluation of a classification pipeline using Azure Face API; received a certificate of completion.
- Built an advisor-selection web system with Java JSP/Servlet and MySQL, covering forms, validation, schema design, persistence, role-based access, preference submission, matching workflow, and CRUD operations.
- Completed a financial-distress warning course paper combining Information Gain/ReliefF with self-organizing map (SOM) pre-selection and a backpropagation (BP) neural-network classifier.

Teaching and Mentoring Experience

Teaching Assistant, UCAS - School of Engineering Sciences

UCAS
2021 - 2023

- Supported Mathematical Methods and Its Applications in Management with 2 TAs, 1 instructor, and **127 students**.
- Supported Accounting and Cost Management with 1 TA, 1 instructor, and **74 students**; supported Human Factors Engineering with 1 TA, 1 instructor, and **14 students**.
- Assisted with tutorials, case discussions, project mentoring, rubrics, homework/exam grading, office hours, logistics, and technical support.

Peer Instructor, BUCT Peer Academic Development Center

BUCT
2018 - 2020

- Delivered peer academic support and course-related instruction for undergraduate students.
- Developed early experience in explaining technical material, organizing peer learning, and supporting course-related problem solving.

Courses Prepared to Teach

- **Data science and information systems:** Data Mining, Machine Learning, Python Programming, Database Systems, Data Visualization, Management Information Systems.
- **Optimization and quantitative decision methods:** Operations Research, Optimization Models and Methods, Decision Analytics, Mathematical Modeling, Combinatorial Optimization.
- **Computing and mathematical foundations:** Programming Fundamentals, Algorithms and Data Structures, Probability and Statistics, Discrete Mathematics, Computational Complexity.



Academic Service, Reviewing, and Program Support

Program Committee Member

AAIM 2026

- 20th International Conference on Algorithmic Aspects in Information and Management.

Peer Reviewer

2023 - Present

- IEEE Transactions on Networking; Expert Systems with Applications; IEEE Transactions on Computational Social Systems; World Wide Web.

Research Seminar and Journal-Club Coordination

UCAS

2020 - 2025

- Presented and discussed research papers in weekly seminars during early doctoral study; later served as paper selector, pre-talk reviewer, and meeting moderator.
- Helped organize weekly research seminars and provided methodological feedback to master's and doctoral researchers.
- Contributed to seminar continuity through agenda preparation, paper screening, discussion moderation, and documentation support.

Graduate Program and Admissions Support

UCAS

2022 - 2023

- Coordinated master's thesis defense and admissions interview processes, including scheduling, documentation, archiving, and other program-support tasks.
- Worked across faculty, students, and administrative timelines to keep defense/admissions processes organized.

Selected Student Leadership and Campus Service

UCAS / BUCT

2017 - 2023

- Served as graduate student union president and class president at UCAS; earlier served as class president and student-organization officer at BUCT.
- Built coordination, communication, and program-support experience across graduate and undergraduate settings.
- Additional campus support included library assistant work at UCAS Yuquan Road Library.



Awards and Honors

- **CAST Doctoral Fellowship:** National Young Talent Support Project, China Association for Science and Technology, national-level program, inaugural cohort, Dec. 2024.
- **National Scholarship:** Graduate, Ministry of Education (China), Dec. 2022.
- **National Scholarship:** Undergraduate, Ministry of Education (China), Dec. 2019 and Nov. 2018.
- **Outstanding Graduate of Beijing:** Beijing Municipal Education Commission, Jul. 2020.
- **Outstanding Young Scholar Paper Award, Second Prize:** International Symposium on Emergency Management, Dec. 2021.
- **University honors:** Merit Student (UCAS, 2022 and 2023; BUCT, 2017); Excellent Student Leader (UCAS, 2022; BUCT, 2018).



Methods, Technical Skills, and Languages

- **Methods:** Combinatorial optimization; complexity and approximation analysis; submodular and non-submodular optimization; sampling and greedy algorithms; graph algorithms; graph neural networks; deep reinforcement learning; Monte Carlo simulation; statistical, ablation, and sensitivity analysis.
- **Programming:** Python (PyTorch, NetworkX, NumPy, pandas), MATLAB, C, Java, and SQL.
- **Data and research tools:** MySQL, SQL Server, MongoDB, Hadoop, SPSS, LaTeX, Git, Linux, and Jupyter.
- **Languages:** Mandarin Chinese (native); English (professional working proficiency).